Results link : results

Code file link(to run these codes we need to pass log-returns based on two assets or 4 assets and we can change the stepsize etc inside the code and can run the files in GPU) :

for horizon 10 and with KDE

for horizon 10 and without kde

for horizon 1 and without kde

for horizon 1 and with kde

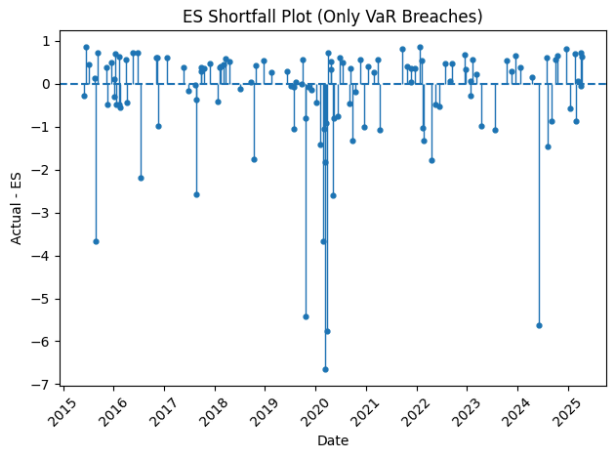
Log returns link : log_4returns

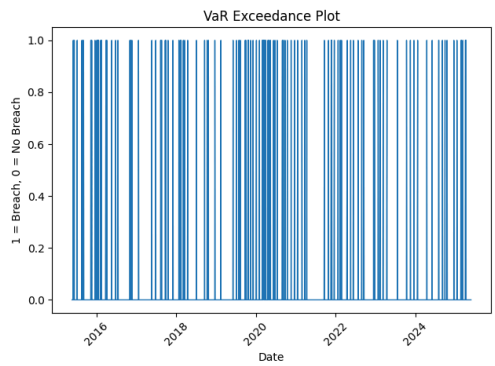
log_returns

If we see the results we can observe that always, with kde model has less mean FZ0 score than no kde model so with kde model is better as our primary evaluation factor is FZ0 score. with no kde model , all backtests gave satisfactory results but in kde model for the case step size 10 , horizon 10 , 4 assets, dq test with lags 4 has p value less than 0.05. So i have increased the simulations to 20k from 10k as you have mentioned and the result is not changing so it is not because of noise .we could also observe that there is not much difference with copula selection based on last 50 or 25 or 100 points . In almost every case fallback occurs less than 15 percent only .

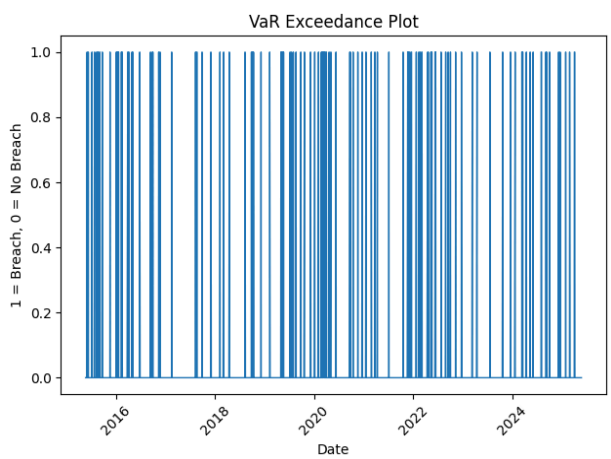
While running above files i have gone through entire code once and i realised one mistake that is, during pit construction i have not normalised KDE , this might be the reason for jump at threshold and that is the reason we are using blending . but now i have normalised the KDE and removed the blending part . I have started running these files after changes, as of now i got some cases results and i am posting in the same excel sheet. now after updating kde normalization where blending is also removed , we can observe that with kde and with no kde model is giving almost same FZo score and also i have checked ks test results(these are also satisfactory) . and also before normalization ,kde model is failing in one test only (for dq test with 4 lags) , in this test implementation degrees of freedom are mentioned wrong , plus 1 should be there so with this change, that dq test error got removed(it is giving satisfactory results after updating it) . we can see that fallback is applied less than 15 percent so the difference comes between with kde and with no kde in less than 15 percent cases only which might be reason for not having much difference.

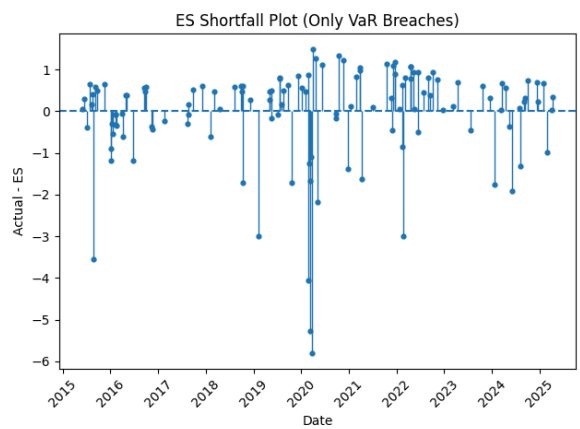
These are the tables before updating KDE normalization :
h=1 s=1 assets = 2 with kde



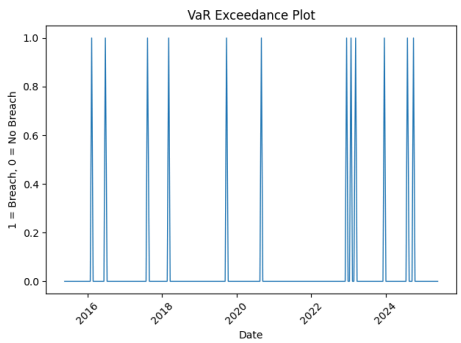


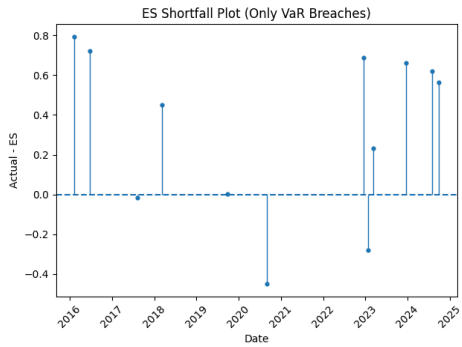
h=1 s=1 assets = 4



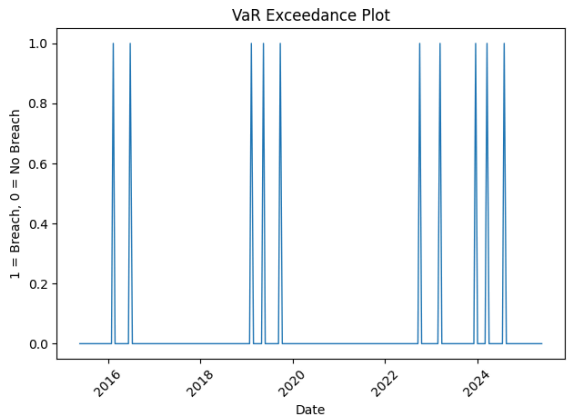


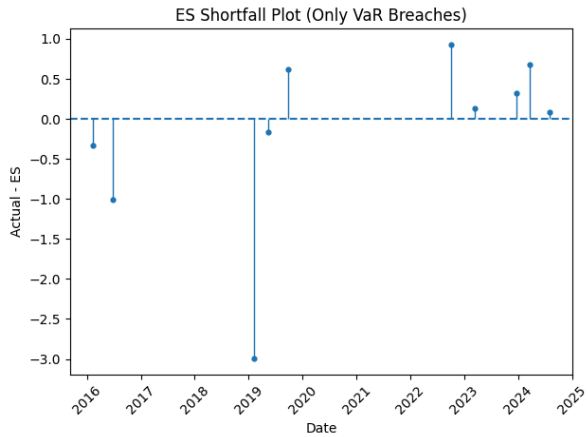
h=1 s=10 assets = 2



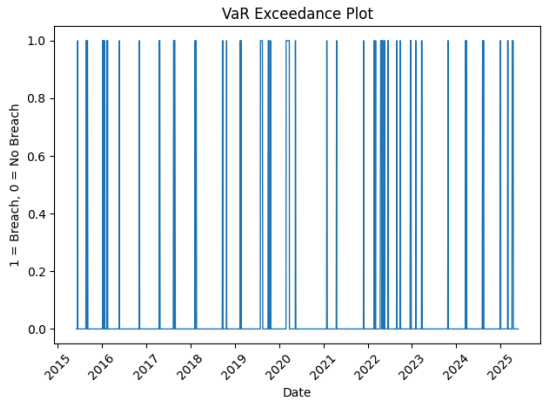


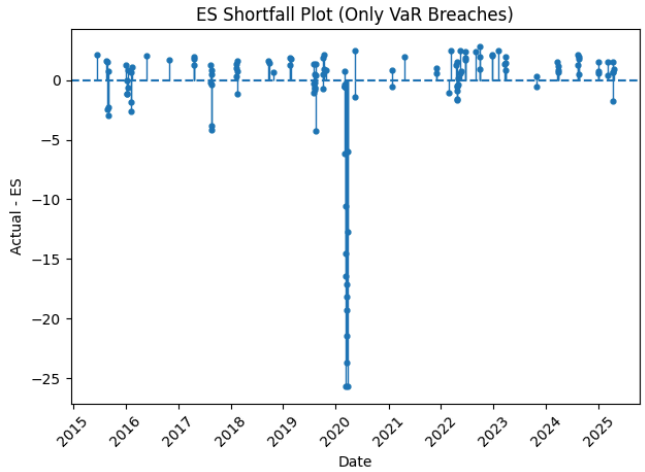
h=1 s=10 assets = 4



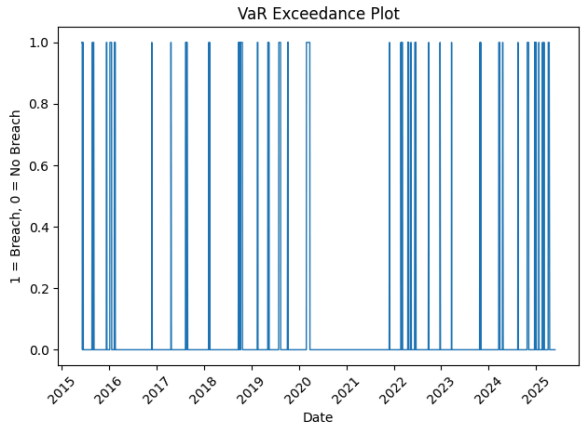


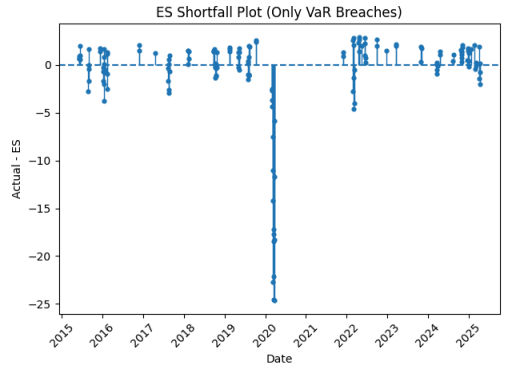
h=10 s=1 assets = 2



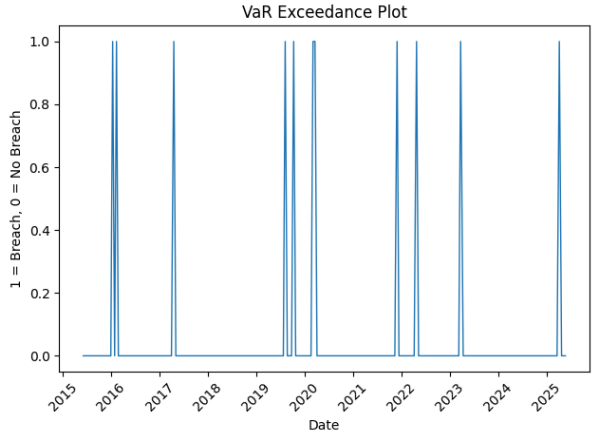


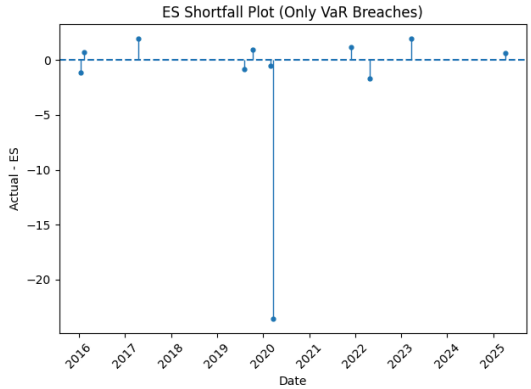
h=10 s=1 assets = 4



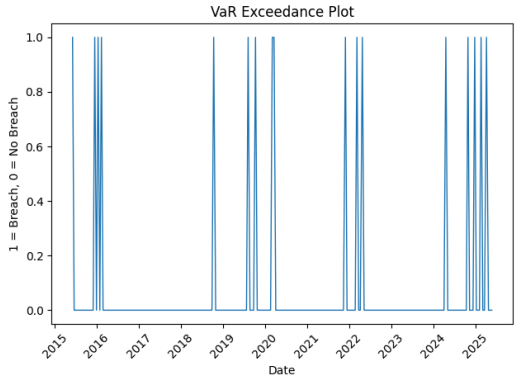


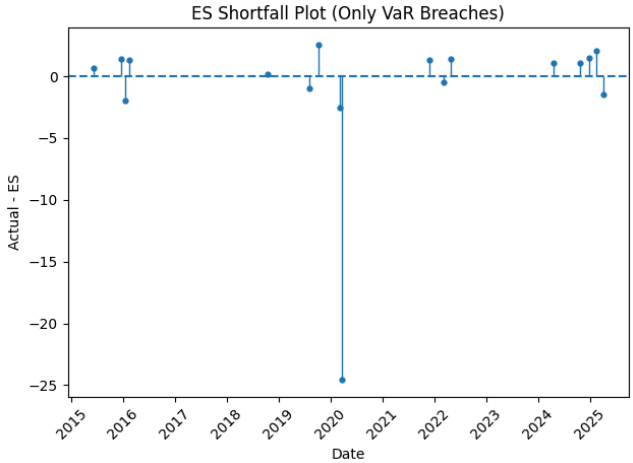
h=10 s=10 assets = 2





h=10 s=10 assets = 4


VaR Exceedance Plot
1 = Breach, 0 = No Breach
1.0
0.8
0.6
0.4
0.2
0.0
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Date


ES Shortfall Plot (Only VaR Breaches)
Actual - ES
0
−5
−10
−15
−20
−25
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
Date